

Lecture 9

Nitrogen, Oxygen and Carbon in the Ocean

Marine Chemistry

Jul 7th 2022

Th 3:00 pm

Guest Lecture: Monica Nelson

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Dissolved gasses in seawater

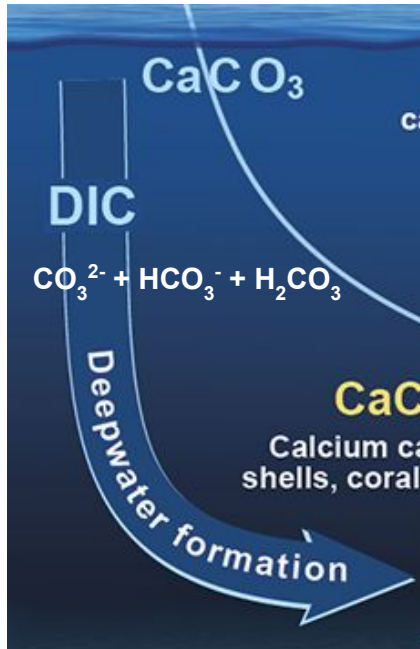
Gas	Percent of Gas in Atmosphere, by Volume
Nitrogen (N ₂)	78.08
Oxygen (O ₂)	20.95
Carbon dioxide (CO ₂)	.0407 in 2018

Garrison. Oceanography.

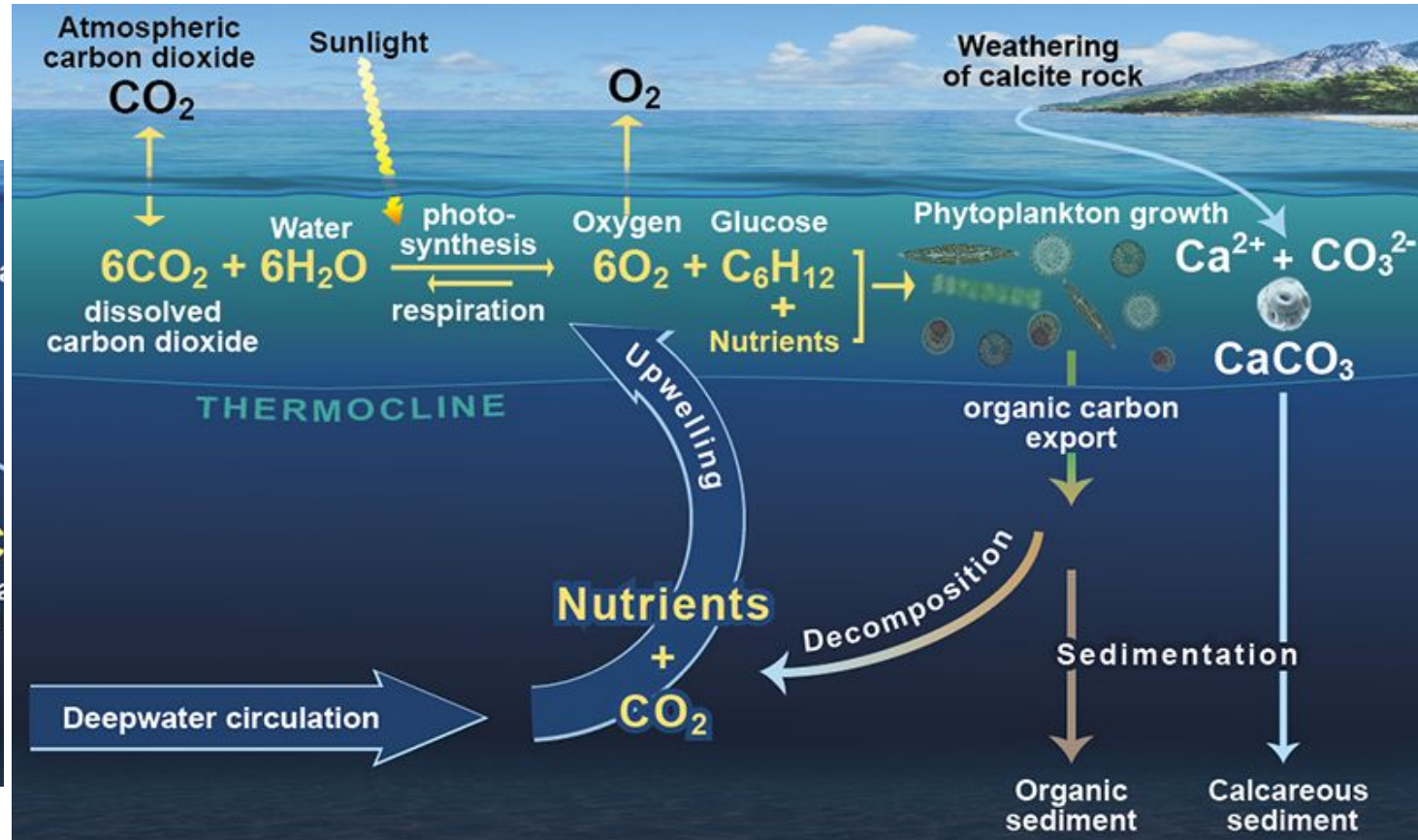
- What are the cycles/processes that govern their concentration?
- How do we measure their concentration?
- What their global distribution and the depth distribution?
- What impact are humans having?

The processes governing these gases are intertwined ...

“Biogeochemical cycle”



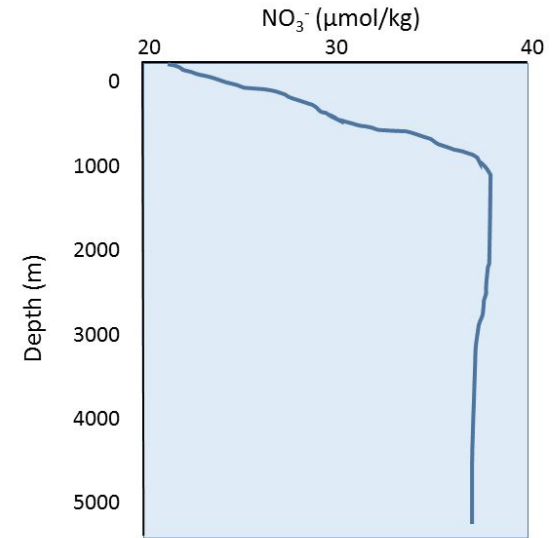
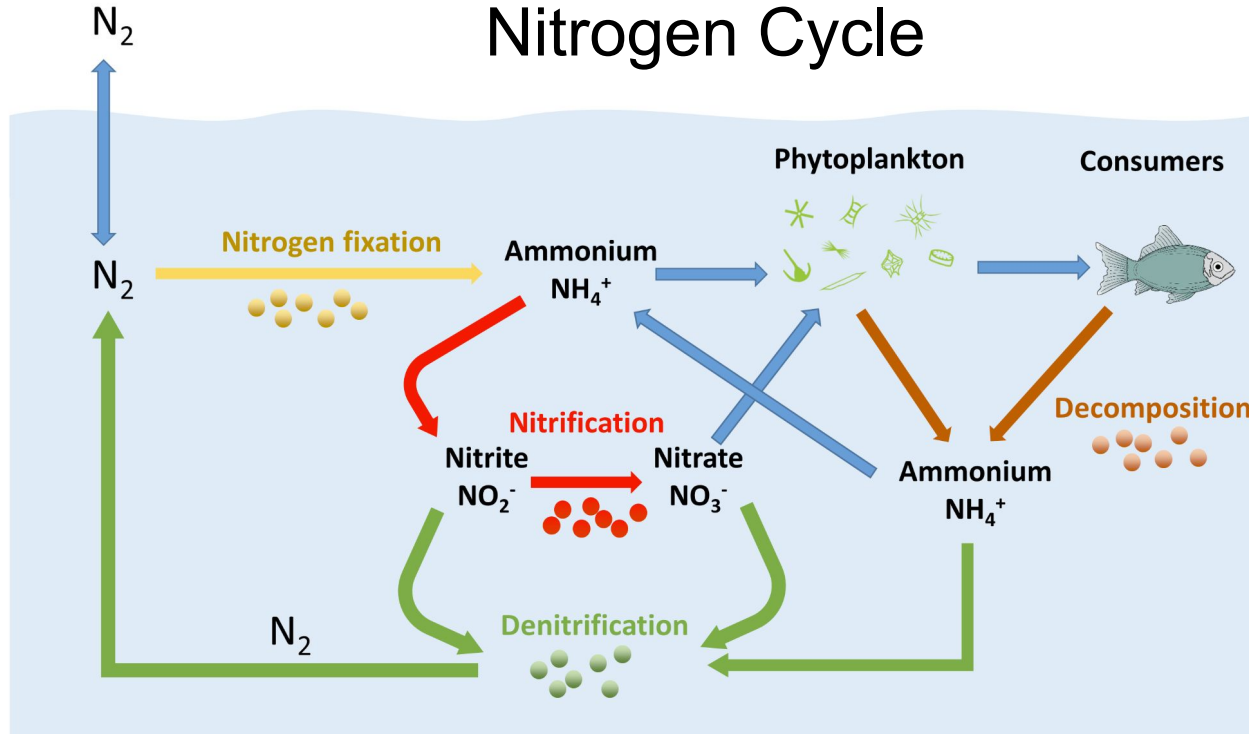
NOC/V.Byfield



Nitrogen - N₂

- Nitrogen is a nutrient that is one of the building blocks of life!
→ Redfield ratio of nutrients in organic material ~ C:N:P = 106:16:1
- When N₂ dissolves in seawater it is inert and unavailable to most life forms
- Special bacteria convert dissolved N₂ into bio-available forms such as Ammonium (NH₄⁺), Nitrite (NO₂⁻), and Nitrate (NO₃⁻)
→ “Nitrogen cycle”

Nitrogen Cycle

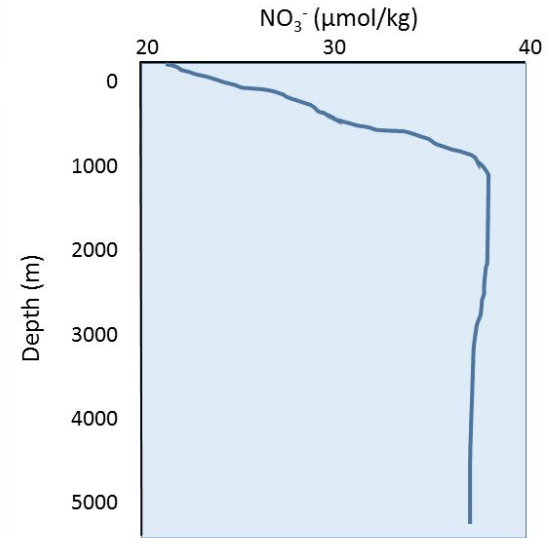
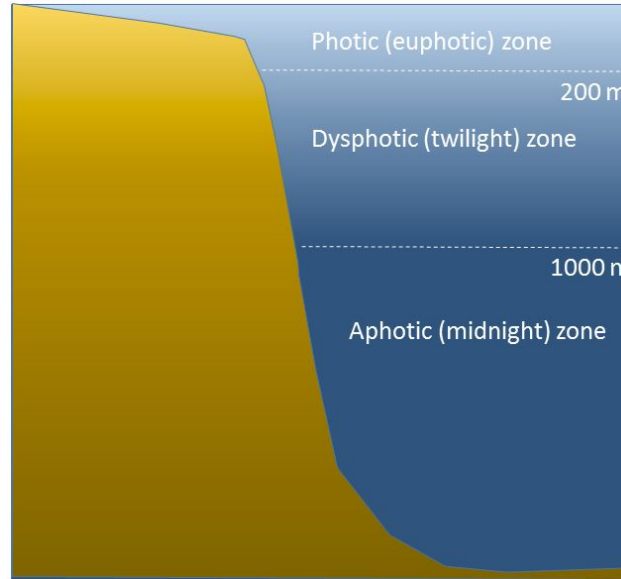


Light Zones

Euphotic zone → enough light to support photosynthesis

Dysphotic “twilight” zone → enough light for predators to see

Aphotic “midnight” zone → less than 1% of incoming light remains

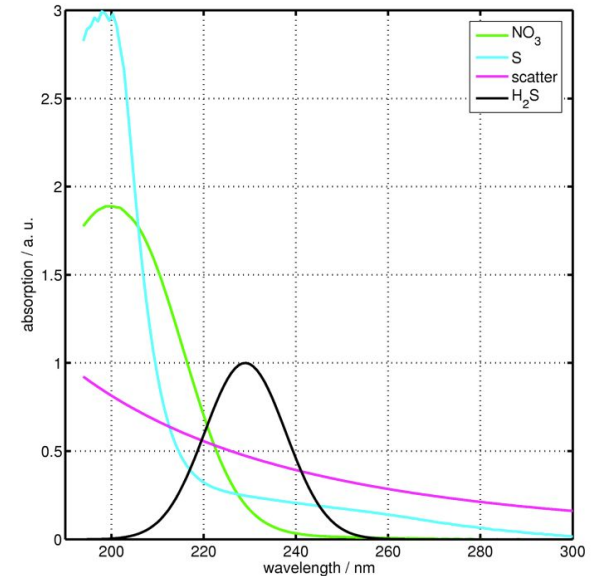


Measuring nitrate

Traditionally, water samples are collected and analysed in a wet lab.

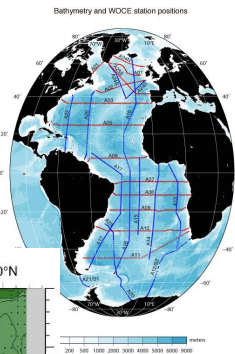
Now, nitrate can be measured optically:

- Nitrate absorbs UV light with a characteristic spectrum that has a maximum at ~200 nm
- By shining UV light through the water and measuring the magnitude of the peak of the absorption spectrum at 200 nm the nitrate concentration can be calculated

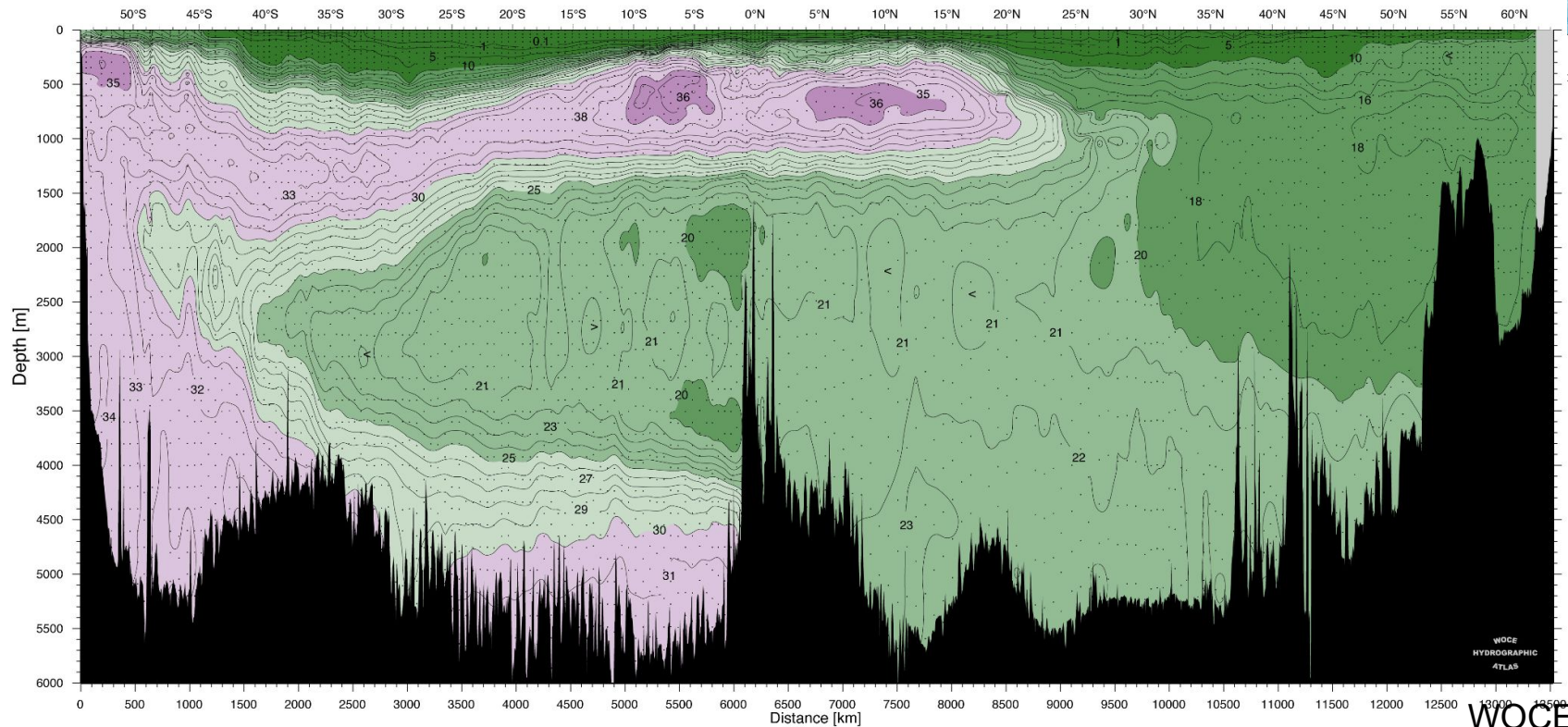


Nitrate distribution

*Do you see a pattern?
What might explain it?*

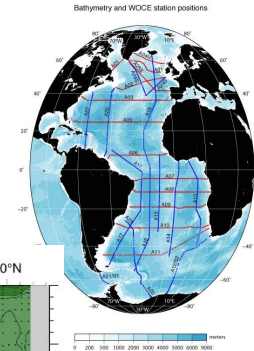


Nitrate [$\mu\text{mol/kg}$] along A16 25 °W

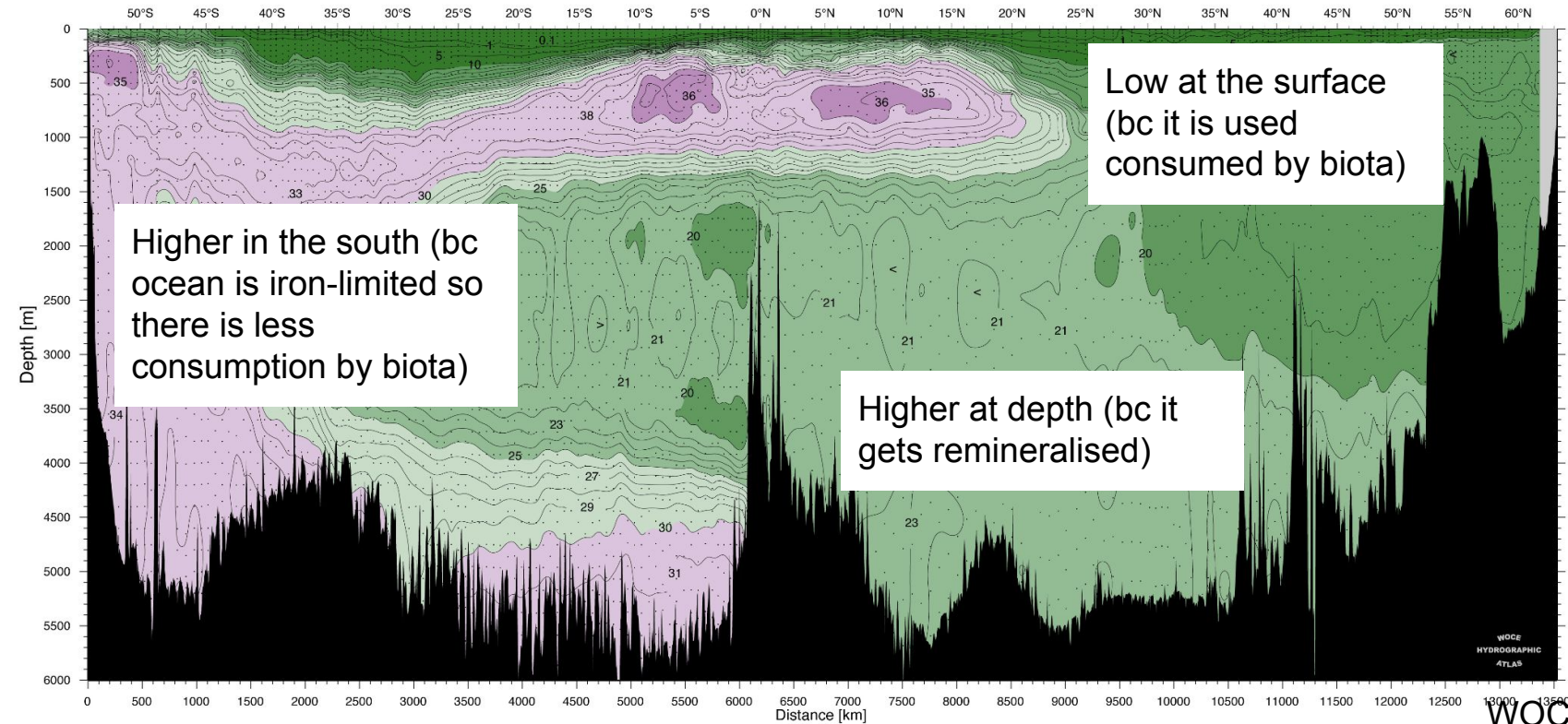


Nitrate distribution

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Nitrate [$\mu\text{mol/kg}$] along A16 25 °W



Human impacts

Eutrophication: When an environment becomes enriched with nutrients (ie nitrate), increasing the amount of plant and algae growth (typically in estuaries and coastal waters).



Causes

- Excess agricultural run-off (fertilizers and animal waste)
- Human sewage and industrial waste

Impacts

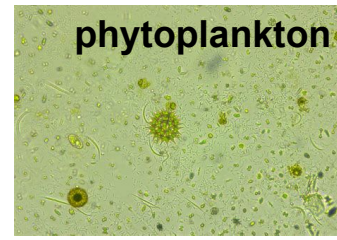
- Algal blooms can sometimes be toxic (Harmful Algal Blooms)
- Decay can lead to low O_2 conditions, fish die-off, and acidification (CO_2 release)

Oxygen - O₂

- Crucial for life
- Important tracer of ocean-atmosphere exchange and interior ocean circulation



Oxygen cycling



phytoplankton and



seaweed

Photosynthesis

VS

Cellular Respiration

Photosynthesis



Respiration

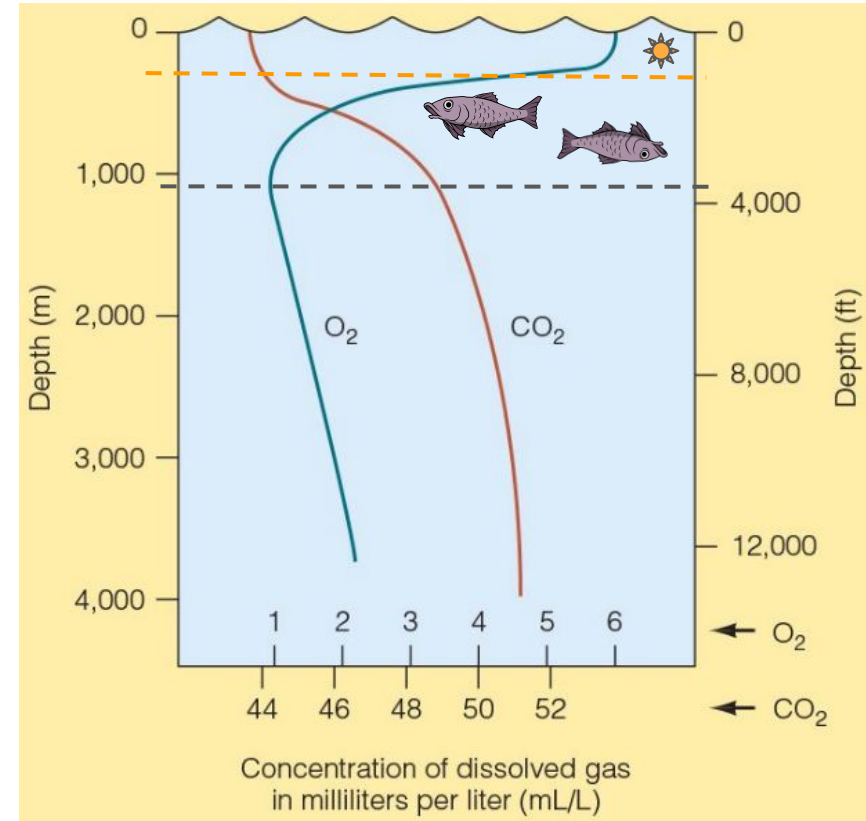


Oxygen distribution with depth

Euphotic zone → high O_2 due to photosynthesis and atmospheric source.

Dysphotic “twilight” zone → low O_2 due to respiration by zooplankton and macrofauna.

Aphotic “midnight” zone → increasing O_2 with depth due to reduced respiration and increasing solubility.



Measuring Oxygen

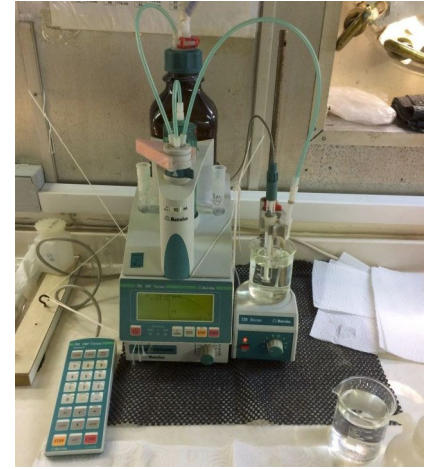
Traditionally:

- Water samples taken back to the lab and titrated using the “Winkler” method
- Method dates back to 1888, but is still the reference standard (with some improvements)!

New optical dissolved oxygen sensors:

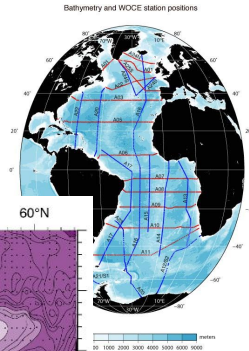
- An oxygen-sensitive sensing foil that containing a substance that when excited with light of a certain wavelength emits light with a different wavelength.
- The amount of O_2 determines the amount of excitation that occurs, which can be quantified based on the phase-shift of the emitted light.

NB: we often still take water samples back to the lab to calibrate the sensor

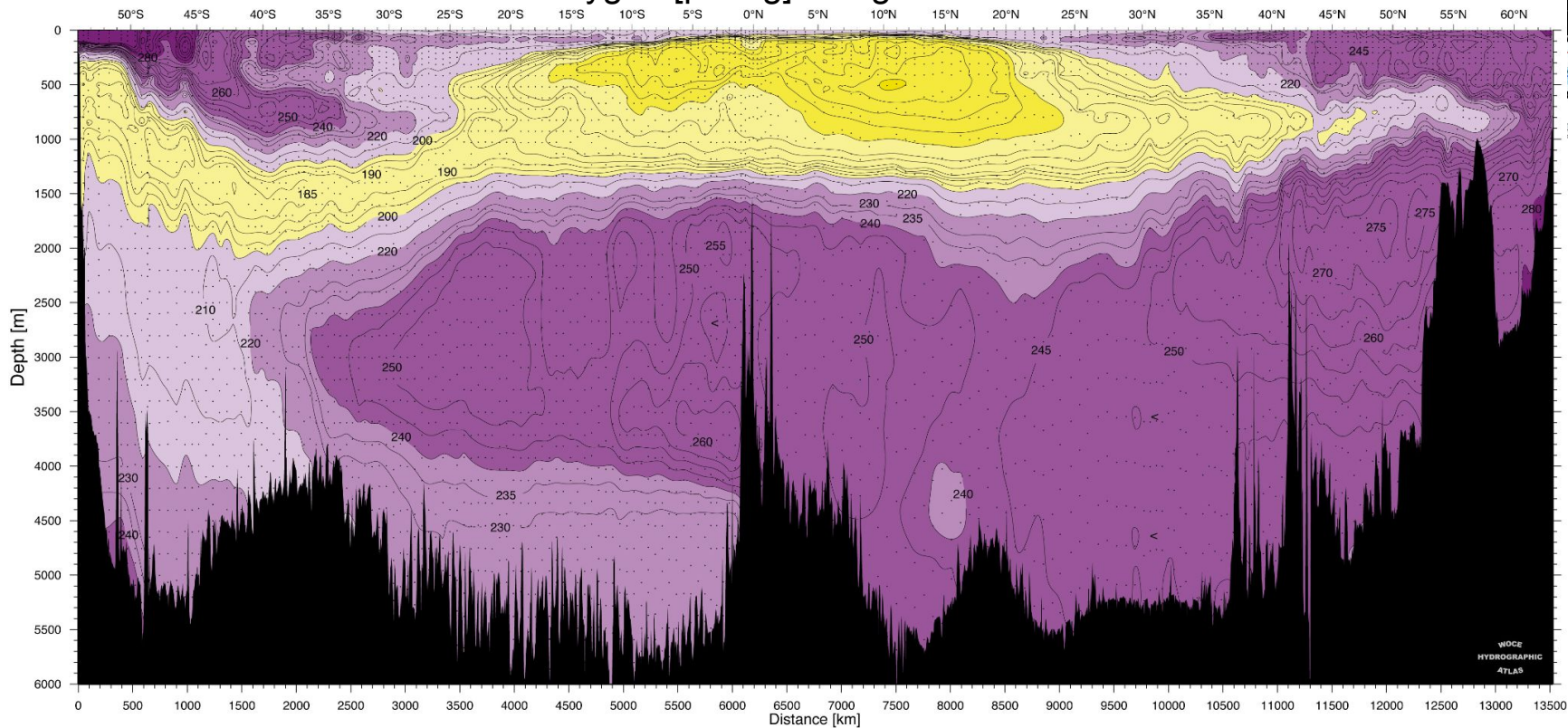


Oxygen Distribution

*Do you see a pattern?
What might explain it?*



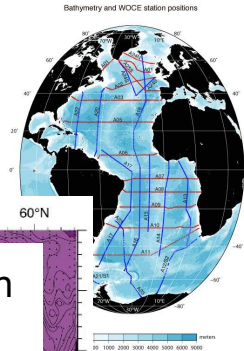
Oxygen [$\mu\text{M/kg}$] along A16 25 °W



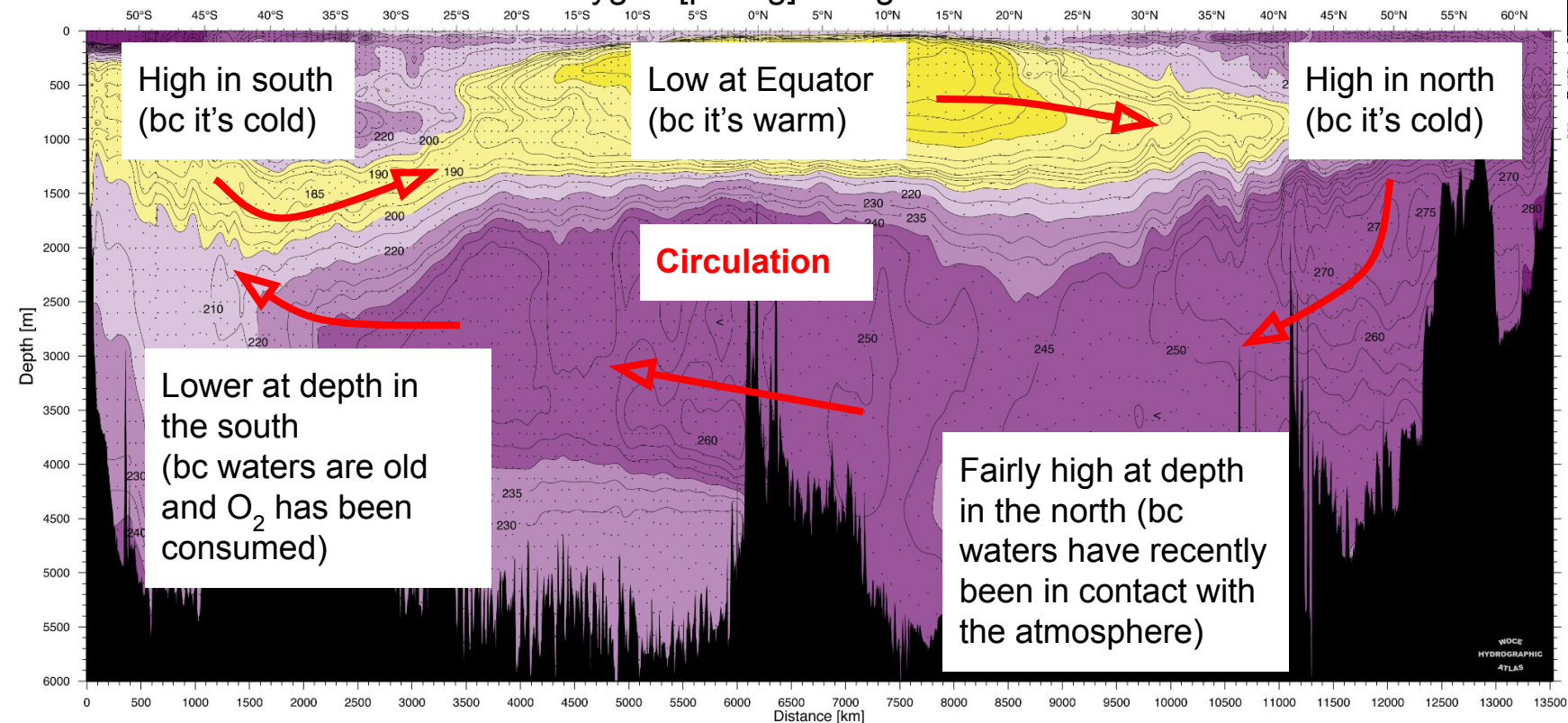
WOCE Atlas

Oxygen Distribution

*Do you see a pattern?
What might explain it?*



Oxygen [$\mu\text{M/kg}$] along A16 25 °W

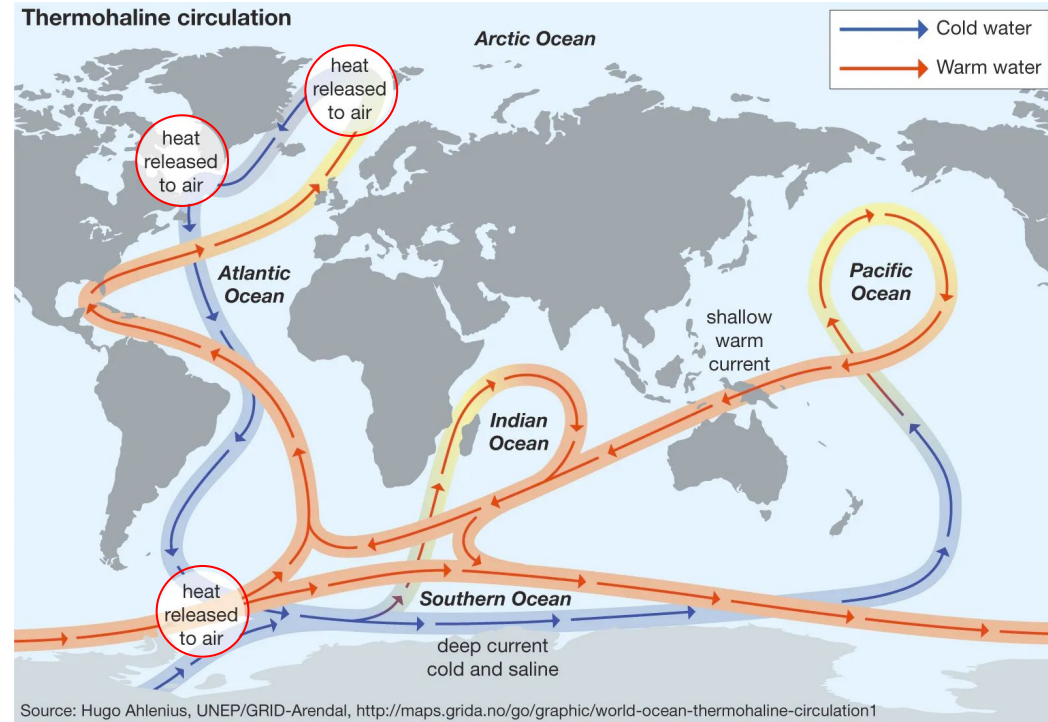


Ocean circulation and ventilation

Ocean ventilation: transport of surface waters into the interior ocean, and in doing so, transporting surface properties to depth.

At high latitudes, when surface waters get very cold and dense they sink to depth, injecting oxygen into deeper waters.

Surface waters in these regions have elevated O_2 because the water is so cold.



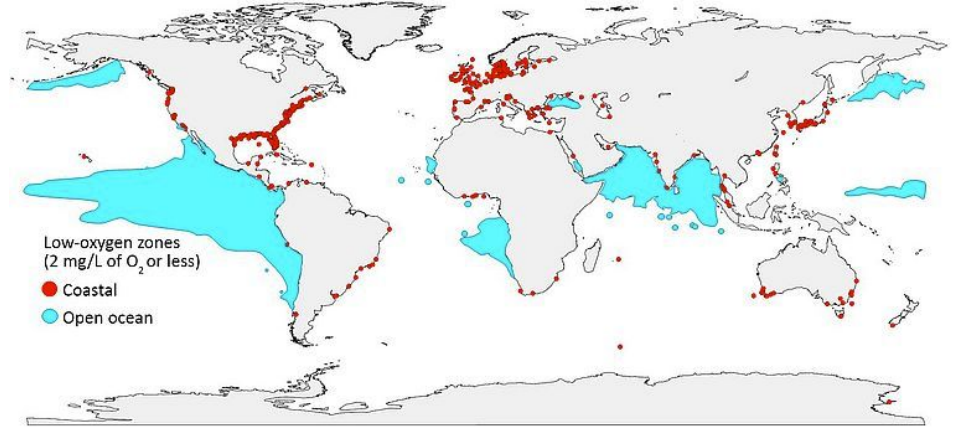
Oxygen Minimum Zones and “Dead Zones”

Hypoxia: a reduced level of oxygen in the water

Anoxia: no oxygen in the water

Human impacts

- Warming oceans → O_2 is less soluble
- Warming oceans → water is more stratified → harder for O_2 -rich waters to be mixed down
- Nutrient-rich pollution (eg. agricultural run-off) → algal blooms that rapidly decompose and use up all the O_2 through respiration, in addition to releasing CO_2

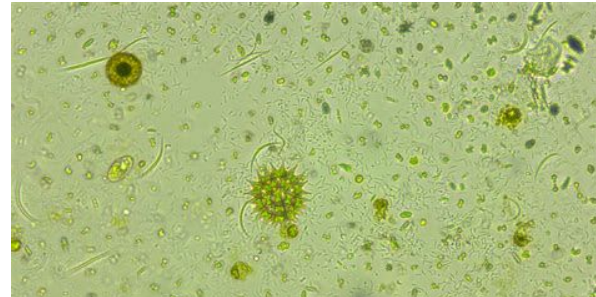
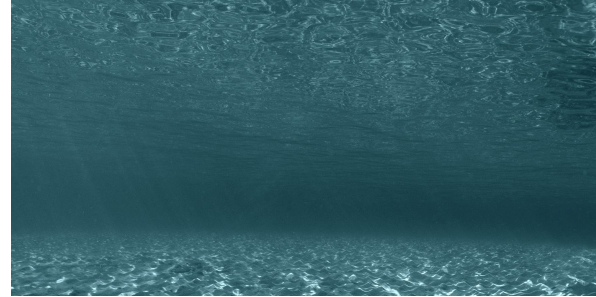


Carbon Dioxide - CO₂

Forms of carbon:

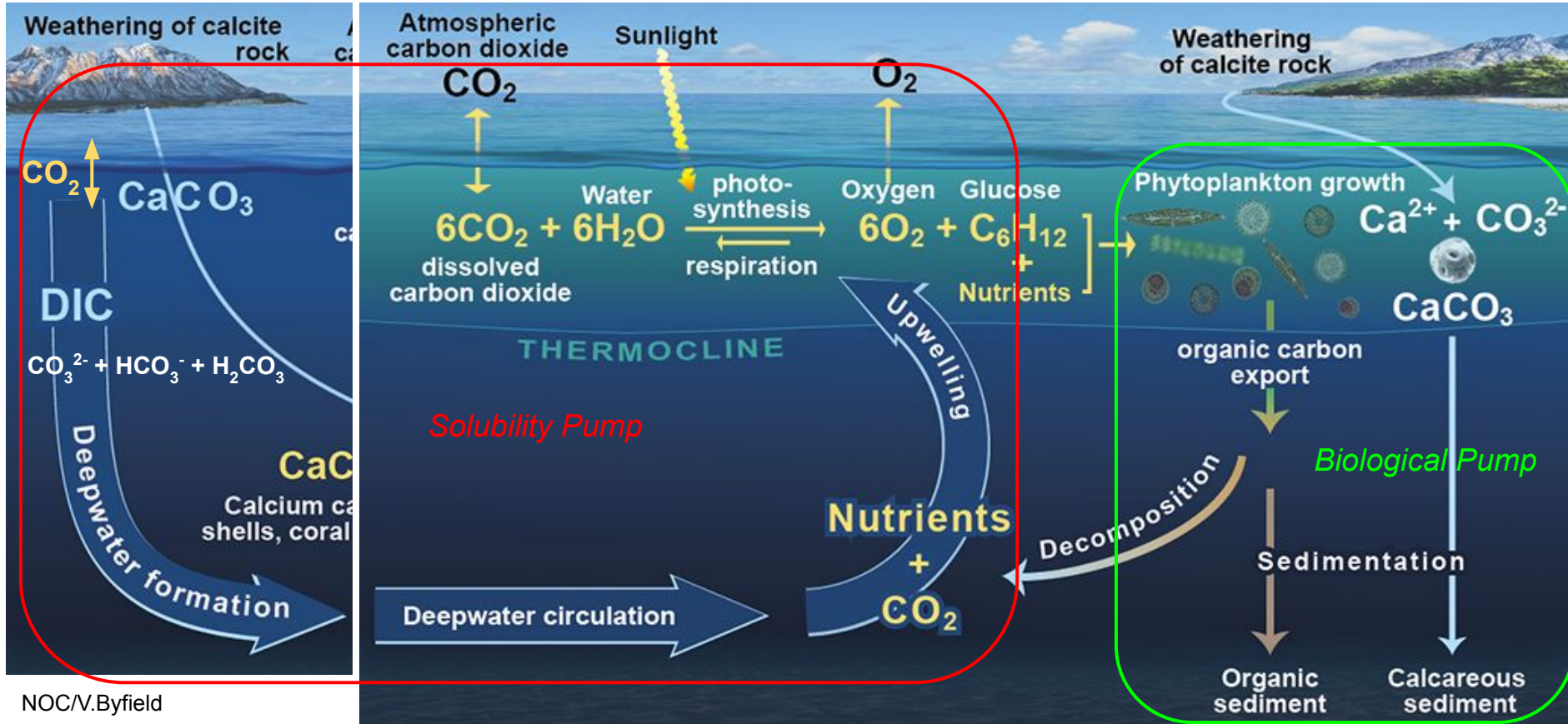
DIC: Dissolved Inorganic Carbon,
ie $\text{DIC} = \text{CO}_3^{2-} + \text{HCO}_3^- + \text{H}_2\text{CO}_3$

POC: Particulate Organic Carbon,
ie small living or dead organisms



Biological and Solubility Pump

NB: It takes about one year to equilibrate CO_2 in the surface ocean with atmospheric CO_2 , so it is not unusual to observe large air-sea differences in CO_2 concentrations. Oxygen equilibrates much faster.



Measuring Carbon

DIC

- Using a coulometer, the amount of DIC in a sample can be determined by the amount of electricity used in electrolysis



POC

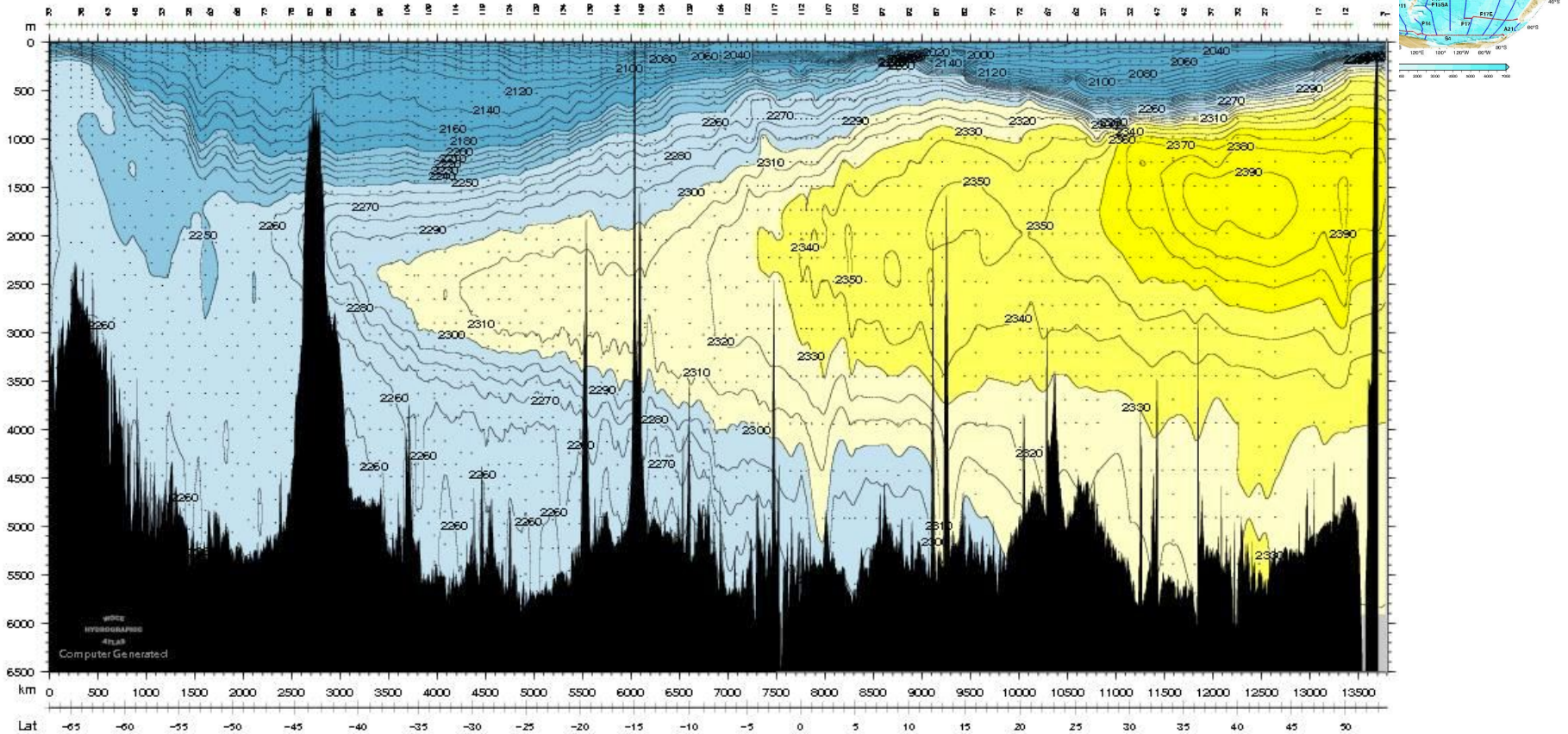
- Can be measured by means of combustion. A water sample is filtered and then the filter containing the POC mass is burnt; the amount of carbon released can be measured.



Carbon Distribution

*Do you see a pattern?
What might explain it?*

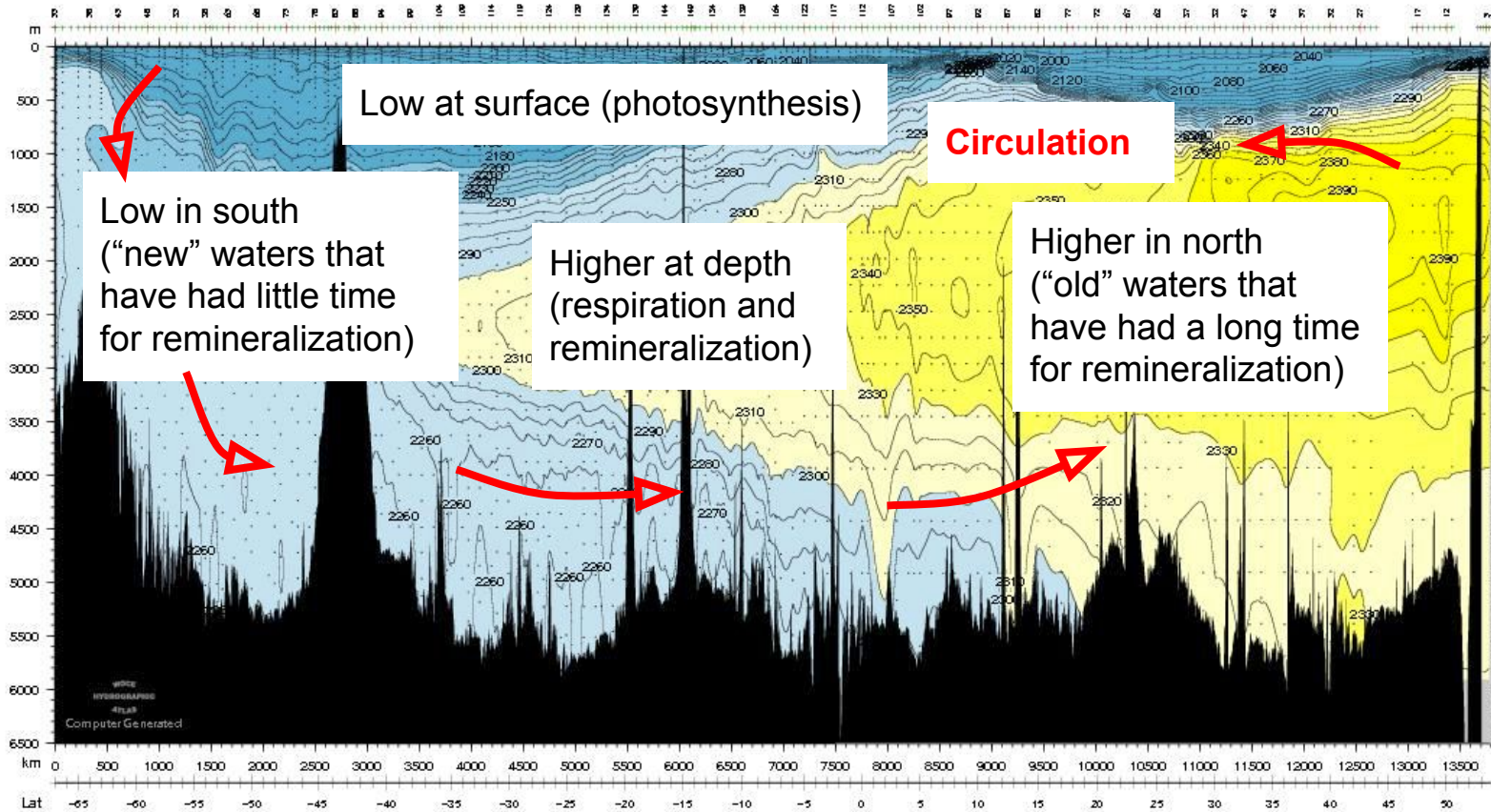
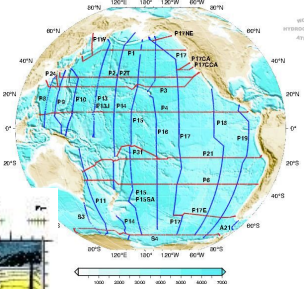
Total CO₂ [$\mu\text{mol/kg}$] for P15 165 °W



Carbon Distribution

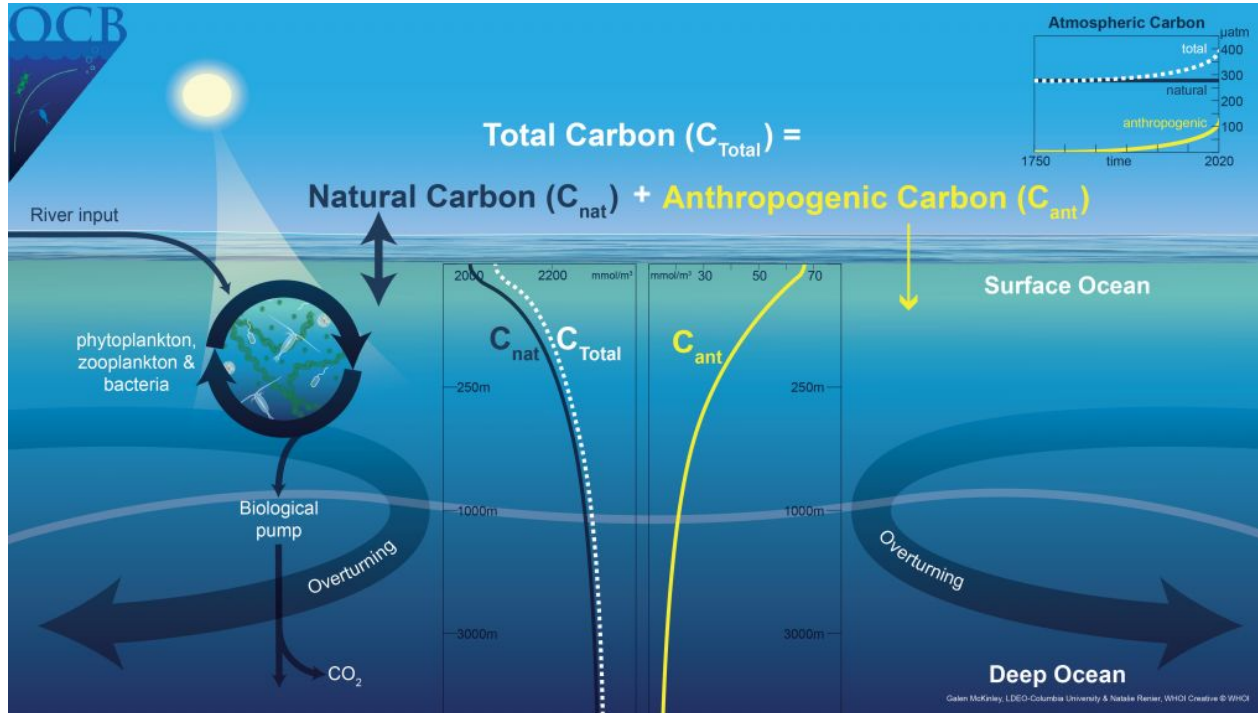
*Do you see a pattern?
What might explain it?*

Total CO₂ [μmol/kg] for P15 165 °W



Human Impacts: Carbon Sequestration and Storage

As atmospheric CO_2 rises, so does oceanic CO_2 .



- Buffers against atmospheric accumulation of CO_2
- Leads to ocean acidification
- As the oceans warm, the oceans' ability to sequester and store carbon may reduce

Human Impacts: Carbon Sequestration and Storage

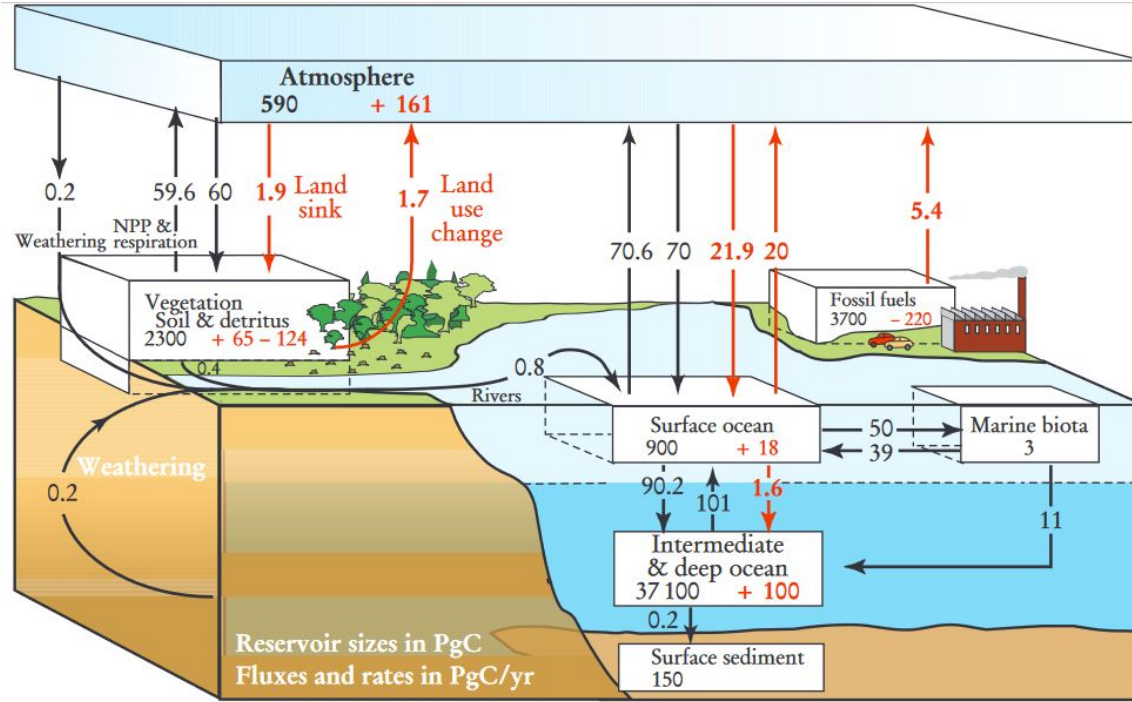


FIGURE 1. GLOBAL CARBON CYCLE. Arrows show the fluxes (in petagrams of carbon per year) between the atmosphere and its two primary sinks, the land and the ocean, averaged over the 1980s. Anthropogenic fluxes are in red; natural fluxes in black. The net flux between reservoirs is balanced for natural processes but not for the anthropogenic fluxes. Within the boxes,

Sarmiento and Gruber. 2002.

Typical NO_3^- , O_2 , DIC profiles in the Pacific and Atlantic oceans

